

Inside an LLM Homework Sheet

Guided homework for Week 2 / Module 0B: Transformers, attention, facts, and verification.

Name

Date

Purpose: These videos deepen the Week 2 classroom examples. Do not try to memorize every architectural detail. Focus on the practical mental model: tokens become representations, attention uses context, learned facts are not the same as verified truth.

Required Video Path

3Blue1Brown: Transformers, the tech behind LLMs

<https://www.youtube.com/watch?v=wjZofJX0v4M>

Watch for: tokens, embeddings, attention, repeated layers, next-token prediction.

3Blue1Brown: How might LLMs store facts?

<https://www.youtube.com/watch?v=9-Jl0dxWQs8>

Watch for: facts in model weights, MLP intuition, why memory is not a source.

Before Watching

In your own words, explain what Week 1 taught: what does an LLM do when it generates an answer?

Write one example where context changes the meaning of a word.

Tiny Glossary

Term	Meaning for this course	Your example
Token	A chunk of text the model processes.	

Embedding	A numeric representation that places a token in a useful meaning space.	
Attention	A mechanism that lets tokens use information from other tokens in context.	
Transformer layer	One repeated block that updates token representations.	
MLP	A component that transforms representations and may help encode patterns or facts.	
Verification	Checking an important claim against evidence outside fluent model output.	

Video 1 Checkpoints: Transformers

Start Tokens and embeddings

Why does the model need to turn text into numerical representations?

Middle Context and attention

Use the word *bank* or *pitcher*. How can surrounding words change the useful meaning?

End Transformer structure

What gets updated again and again inside the transformer?

Video 2 Checkpoints: Facts

Start Facts in a model

What does it mean to say a fact may be encoded in model weights?

Middle MLP intuition

What role do MLP layers appear to play in the video's explanation?

End Memory vs evidence

Why is learned factual recall different from checking a source?

Fact or Source?

Claim	Use model recall, source, tool, or human approval?	Why?
The Eiffel Tower is in Paris.		
Our class policy allows late work after illness.		
The library closes at 6pm today.		
A private class decision from yesterday.		

Exit Ticket

In 5-7 sentences, explain the full Week 2 mental model: text -> tokens -> embeddings -> attention/context -> repeated layers -> prediction -> verification.

Completion Checklist

- ☐ I watched the transformer video and answered the checkpoint questions.
- ☐ I watched the facts video and answered the checkpoint questions.
- ☐ I completed the glossary with my own examples.
- ☐ I can explain why learned facts still need verification in important cases.

Teacher Grading Guide

Criteria	Points
Accurate glossary with examples	20
Transformer video checkpoints show understanding of embeddings and attention	25
Facts video checkpoints distinguish learned memory from evidence	25
Fact/source table uses appropriate verification judgment	15
Exit ticket connects the whole flow clearly	15